Classical Field Theory — Exercise Sheet 16

Energy, plateaux and local blow-up

Hints and answers

Use the general one-dimensional local theorem proved on Sheet 15: for $V \in C^3$, $V'(0) = 0$, data in $H^1(\mathbb{R}) \times L^2(\mathbb{R})$ have a unique weak energy solution, and a finite maximal endpoint forces its full energy-space norm to tend to infinity. Its reference linear mass is one. The conservation argument below must be made at that weak regularity.

Exercise 16.1 (A nonnegative potential beyond the quartic). Let $V \in C^3(\mathbb{R})$ satisfy $V \geq 0$ and $V(0) = V'(0) = 0$. For the energy solution of $u_{tt} - u_{xx} + V'(u) = 0$ with data (u_0, g_0), set

$$E(t) = \int_{\mathbb{R}} \left[\frac{1}{2} u_t^2 + \frac{1}{2} u_x^2 + V(u) \right] \mathrm{d}x.$$

(a) Prove finiteness, continuity and conservation of E by smoothing the actual solution in space. Justify the potential chain rule and the endpoint limits. Do not assume a density theorem for nonlinear solutions.

(b) Prove global existence and give a finite-time bound on the full $H^1 \times L^2$ norm. For $V(s) = s^2/2 + s^6/6$, obtain a stronger bound uniform in time. Identify which part of the proof needs $V \geq 0$.

Hint. On a compact lifespan interval, $H^1(\mathbb{R})$ bounds the range of u. Use $|V(a) - V(b)| \leq C_M(|a| + |b|)|a - b|$ there. For a fixed smoothing parameter, Taylor's remainder is quadratic in the L^2 time increment; the smoothed field has enough norm derivatives.

Answer. Energy is constant and

$$\|(u(t), u_t(t))\|_{\mathrm{lin}}^2 \leq \left(\|u_0\|_2 + |t| \sqrt{2E_0} \right)^2 + 4E_0.$$

For the stated sextic potential the stronger bound is $\|(u(t), u_t(t))\|_{\mathrm{lin}}^2 \leq 2E_0$.

Exercise 16.2 (Sine-Gordon energy does not bound every phase norm). Take $V(u) = 1 - \cos u$ and background zero.

(a) Use Problem 1 to prove energy conservation and global energy-class evolution. Verify explicitly the initial potential integrability and continuity needed for that argument.

(b) Choose a fixed nondecreasing $\eta \in C^\infty(\mathbb{R})$, zero for $s \leq 0$ and one for $s \geq 1$. For $R \geq 1$ take initial displacement

$$u_R(x) = 2\pi\eta(x + R + 1)\eta(R + 1 - x)$$

and zero velocity. Compute its energy in terms of two integrals on $[0, 1]$ and show that it is independent of R, while $\|u_R\|_2 \to \infty$. Does this contradict global existence or the finite-time bound of Problem 1?

Hint. Use $0 \leq 1 - \cos s \leq s^2/2$ and $|\sin s| \leq |s|$. The plateau has zero potential, but its two fixed transition ramps do not.

Answer.

$$E[u_R, 0] = 4\pi^2 \int_0^1 \eta'(s)^2 \mathrm{d}s + 2\int_0^1 [1 - \cos(2\pi\eta(s))]\mathrm{d}s, \qquad \|u_R\|_2^2 \geq 8\pi^2 R.$$

Each datum evolves globally, but this fixed energy does not uniformly control the displacement norm of the entire family.

Exercise 16.3 (Focusing blow-up from compact data on the line). Change the potential to $V(u) = -u^4/4$ on $\mathbb{R}$, so $u_{tt} - u_{xx} = u^3$. Fix $0 < T < R$ and choose a real $\chi \in C_c^\infty(\mathbb{R})$ with $0 \leq \chi \leq 1$ and $\chi = 1$ on $[-R, R]$. Take

$$u_0 = \frac{\sqrt{2}}{T}\chi, \qquad g_0 = \frac{\sqrt{2}}{T^2}\chi.$$

For this conditional exercise, assume additionally that the unique energy solution for the displayed data has a C^2 representative throughout its maximal energy-class lifespan $[0, T_*)$. This is a regularity hypothesis, and assumes no global existence.

(a) Compute its finite initial focusing energy. Compare on the cone $|x| \leq R - t$ with $q(t) = \sqrt{2}/(T - t)$ for every common time $0 \leq t < \min\{T, T_*\}$. Prove the needed local uniqueness rather than assuming a global finite-energy theorem for q.

(b) Deduce $T_* \leq T$. State carefully what was proved if the maximal solution terminates earlier than T. Why do these data improve on simply placing the spatially constant ODE on all of $\mathbb{R}$?

Hint. For $w = u - q$, factor $u^3 - q^3$ and use a positive mass-one energy on the shrinking interval. Its derivative is at most a bounded coefficient times that energy; zero initial energy then stays zero.

Answer.

$$E_0 = T^{-2}\int \chi'^2 \mathrm{d}x + T^{-4}\int (\chi^2 - \chi^4)\mathrm{d}x > 0.$$

On the common cone $u = q$, so a solution extending past T is impossible. The maximal energy lifespan is finite, with $T_* \leq T$.